SECTION 04 · CLUSTERING

Clustering Digits and Segmenting Customers

Find groups with no labels: centroids, hierarchies, density and mixtures, on MNIST and on 440 real customers.

k-means | HDBSCAN | Silhouette | Dendrogram | DBSCAN | Gaussian Mixture

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Laxmi Kant Tiwari · kgptalkie.com

What Clustering Does



Centroids

`k-means`

nearest centre wins

Hierarchy

`Ward linkage`

merge closest groups

Density

`DBSCAN, HDBSCAN`

crowded regions, rest is noise

Mixtures

`Gaussian mixture`

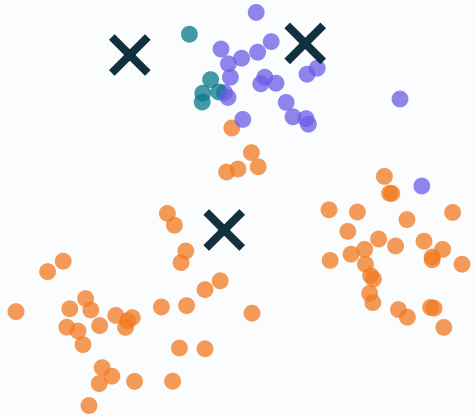
a probability per cluster

k-Means Step by Step

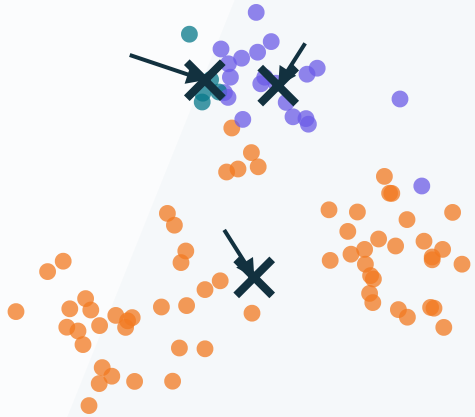
1. drop k centres



2. assign to nearest



3. move to the mean



4. repeat until still



Inertia: the sum of squared distances from each point to its centre. Each step lowers it.

The result depends on the start, so KMeans tries several starts and keeps the best.

What k-Means Assumes

k-means on two half moons



it cuts straight across both shapes

k-means works best when clusters are

- round, like balls around a centre
- about the same size and spread
- on features with comparable scales

Other shapes need other ideas

density methods follow curved, uneven groups; mixtures allow stretched ellipses

How to Grade Clusters

clusters vs held-back labels

	label 0	label 1	label 2
cluster A	**40**	3	2
cluster B	5	**30**	20
cluster C	2	4	**35**

purity = sum of the highlighted counts / all points

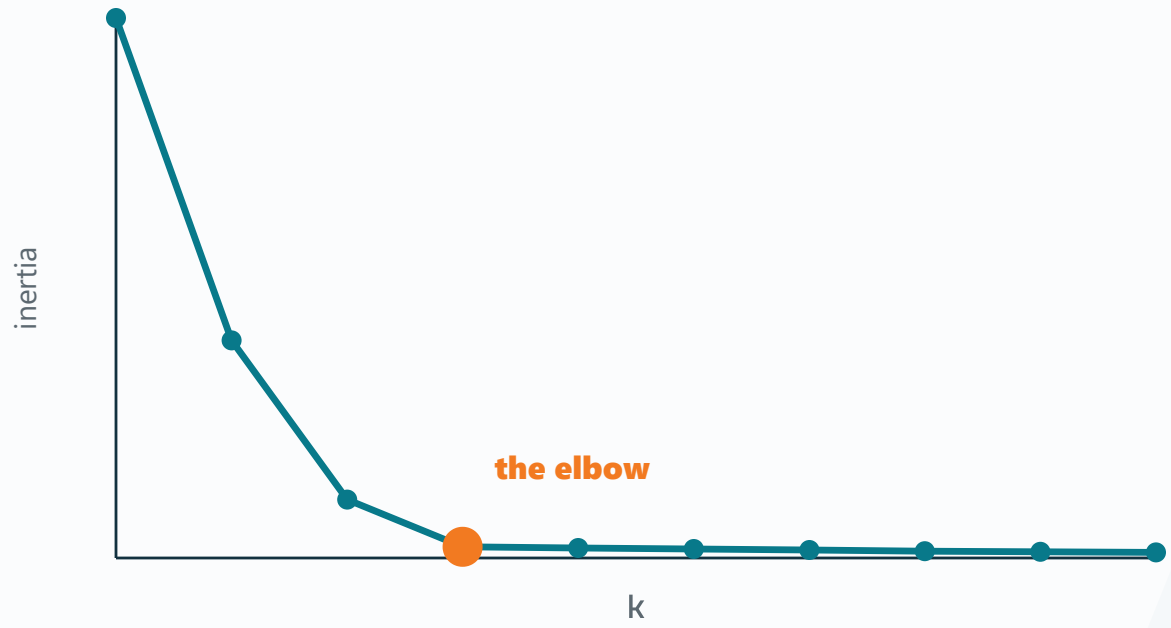
Purity — label each cluster by its majority
rises with k: one point per cluster scores 1

Majority baseline — everything in one cluster
the score to beat

Adjusted Rand index — do pairs of points agree?
0 for chance, 1 for a perfect match, any k

Silhouette — no labels at all
how far each point sits from the next cluster

The Elbow Method



Inertia always falls as k grows

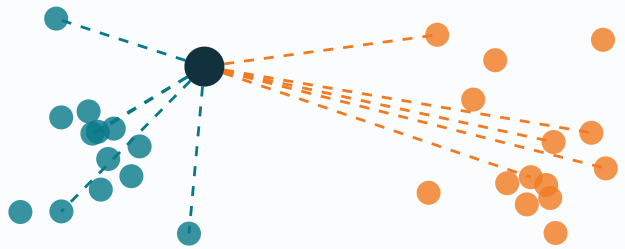
every new centre sits closer to someone

Look for the bend: after it, extra clusters buy very little

Real data rarely has a sharp elbow

on digits the curve bends smoothly, so pair it with silhouette

The Silhouette Score



a = mean distance to its own cluster

b = mean distance to the nearest other cluster

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silhouette = (b - a) / max(a, b)
```

near 1: deep inside its cluster

near 0: on the border

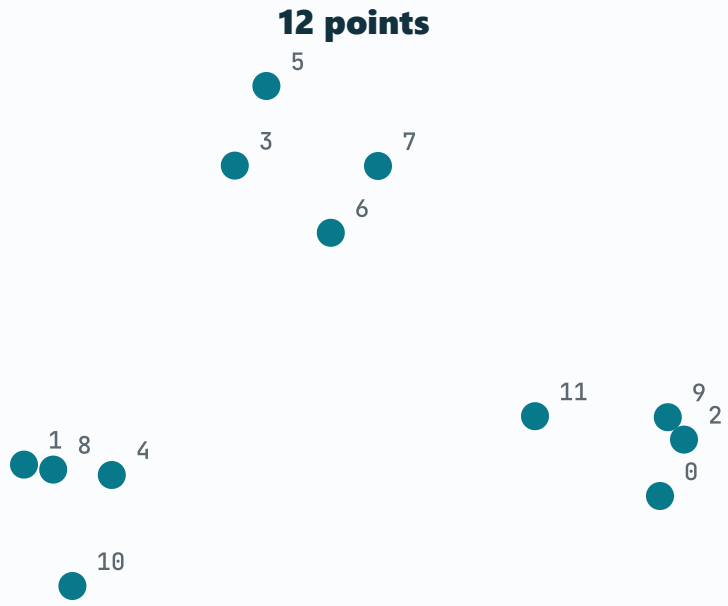
below 0: closer to another cluster

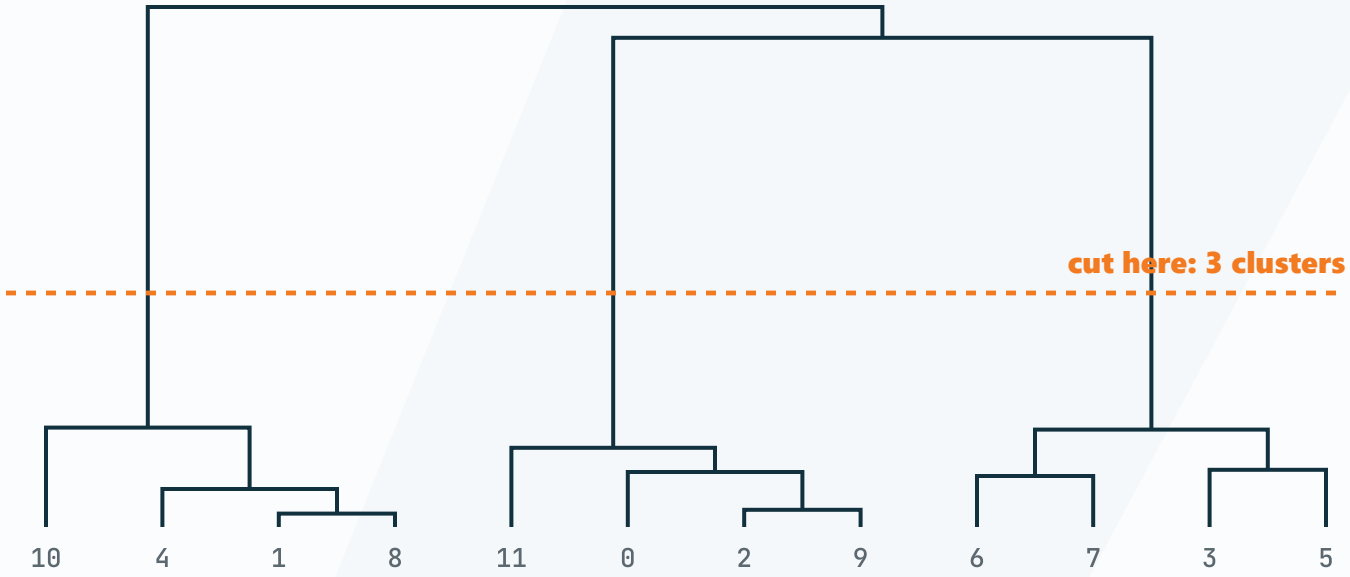
Average over all points

pick the k with the highest average;

it costs one distance per pair of points

Hierarchical Clustering and the Dendrogram

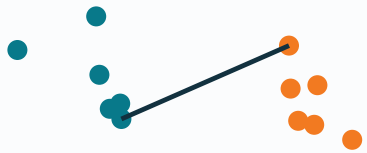




height of a join = how far apart the two groups were

Single, Complete, Average and Ward Linkage

single
closest pair



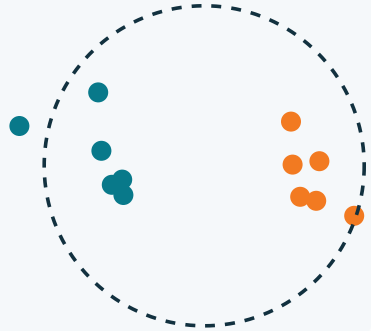
complete
farthest pair



average
mean of all pairs

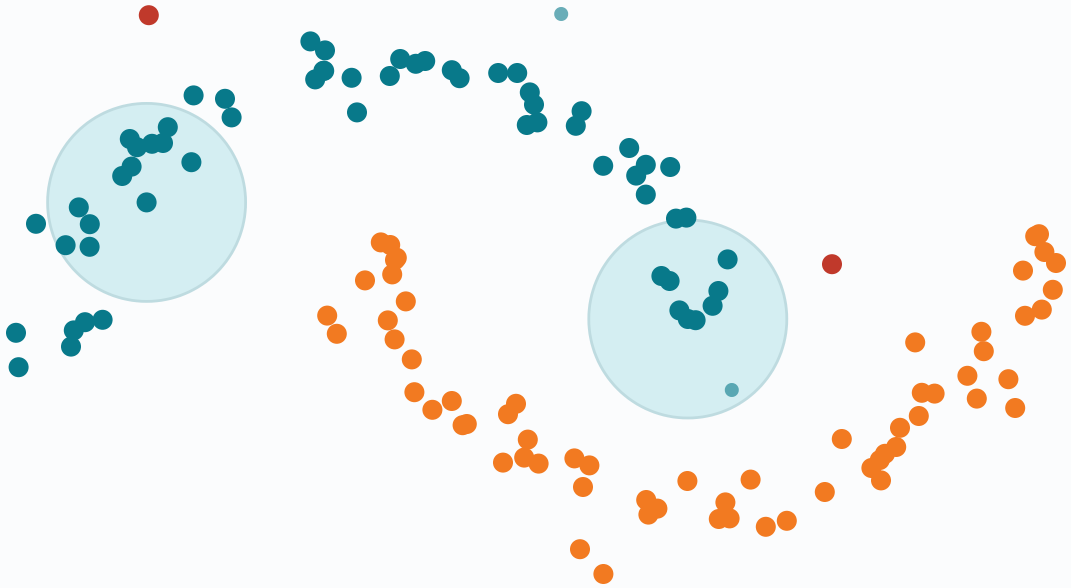


ward
smallest rise in spread



Ward is the usual choice for compact, similar-size clusters
single linkage chains through thin bridges; complete and average sit in between

How DBSCAN Grows Clusters



Two settings:

- eps: the radius of a neighbourhood
- min_samples: points needed inside it

core point: crowded neighbourhood

border point: next to a core point

noise: neither, labelled -1

Clusters grow from core to core

any shape, and k is found, not given

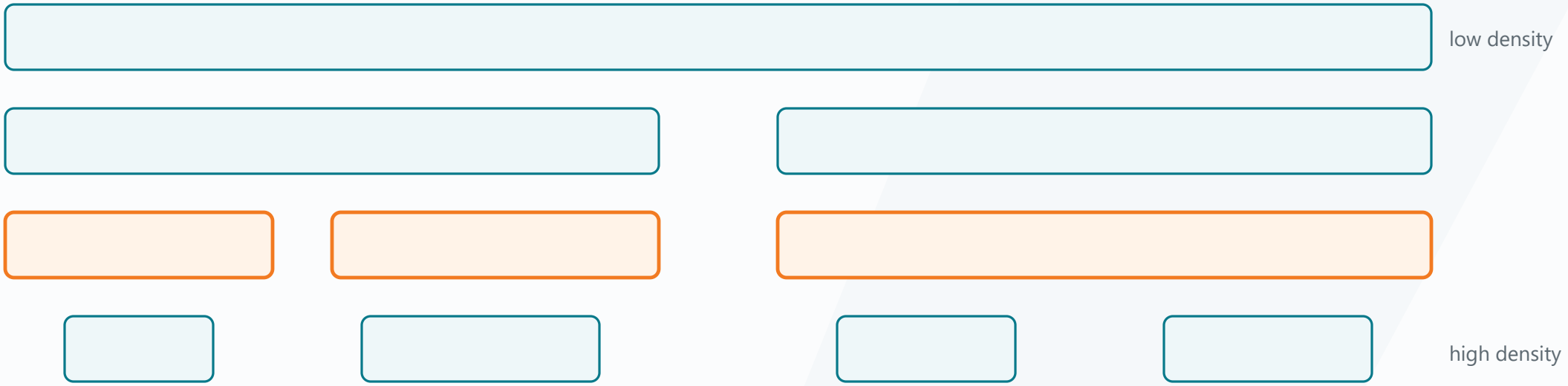
Why eps Is Hard to Choose



One radius cannot suit a tight group and a loose group at the same time.

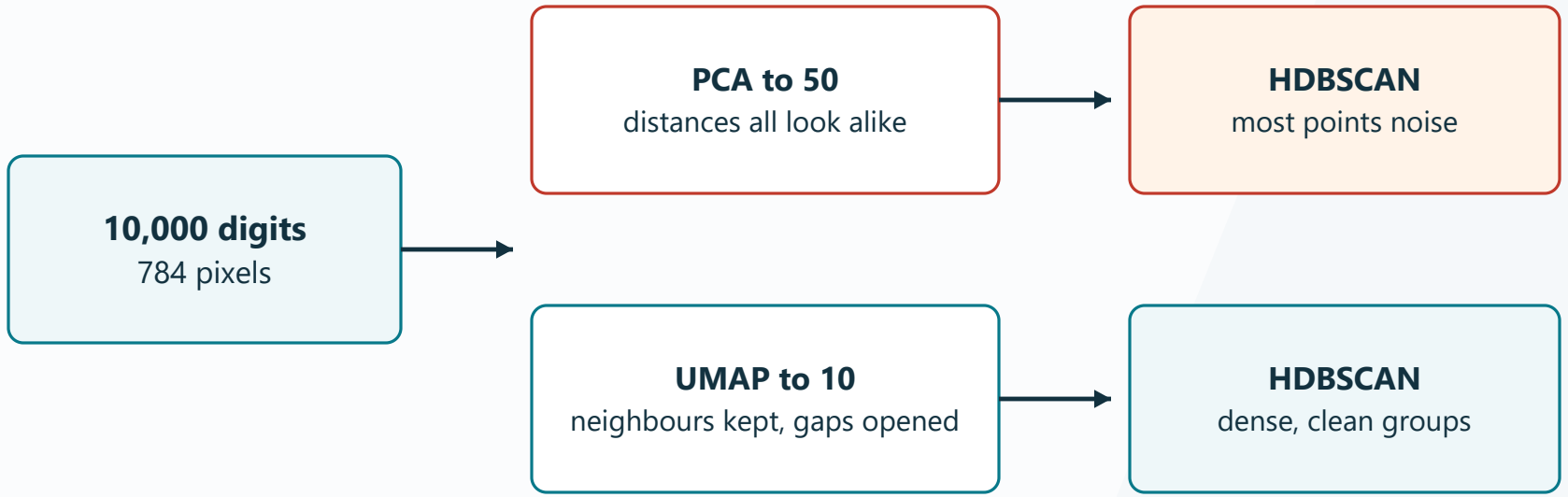
HDBSCAN Removes eps

Every eps at once: a tree of clusters as the density threshold rises



HDBSCAN keeps the branches that last longest (orange); min_cluster_size drops tiny ones

Density Clustering Needs Fewer Dimensions

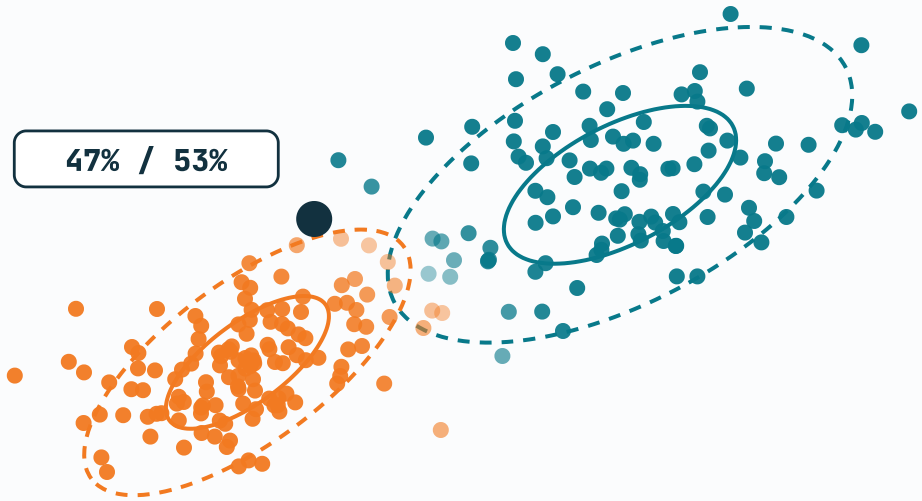


Groups it finds are real but not always the labels:

4, 7 and 9 are written alike and can share one cluster

Density needs meaningful distances. Reduce with a neighbour-preserving method first.

Soft Clusters With Gaussian Mixtures



Each cluster is a Gaussian ellipse with its own centre, spread and tilt

predict: the most likely cluster (hard)

predict_proba: a share for every cluster (soft)

Soft memberships flag the uncertain
points between clusters, the customers worth a second look

The Wholesale Customers Data

yearly spend per product group

- Fresh
- Milk
- Grocery
- Frozen
- Detergents_Paper
- Delicassen

- Channel — held back to grade

440 customers of one wholesaler

Channel 1: hotels, restaurants, cafés
Channel 2: retail shops

The question a sales team asks
which groups of customers buy alike,
and what should each group be offered?

Why Scaling Changes Clusters

raw spend



a few big spenders stretch the axes

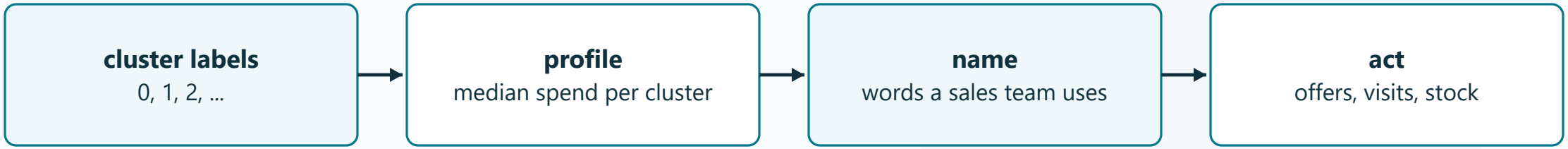
log of spend



spread evenly, groups visible

Clusters follow distances, and distances follow units and skew: log, then StandardScaler.

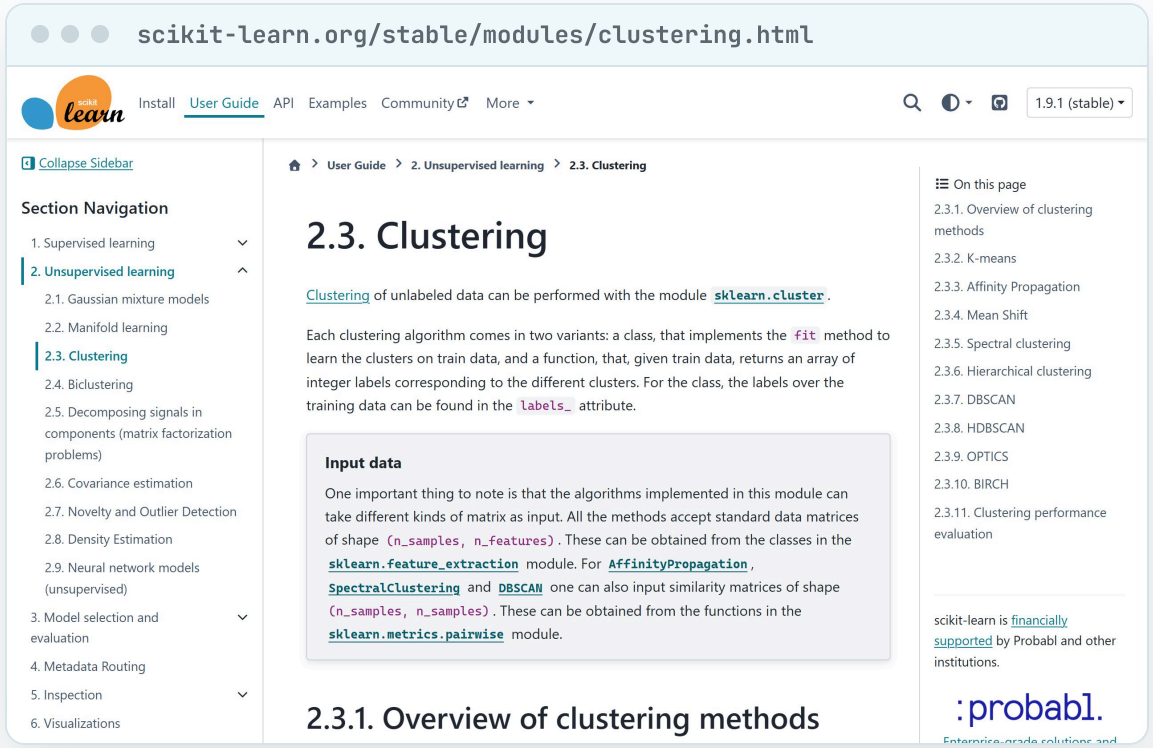
From Clusters to Customer Segments



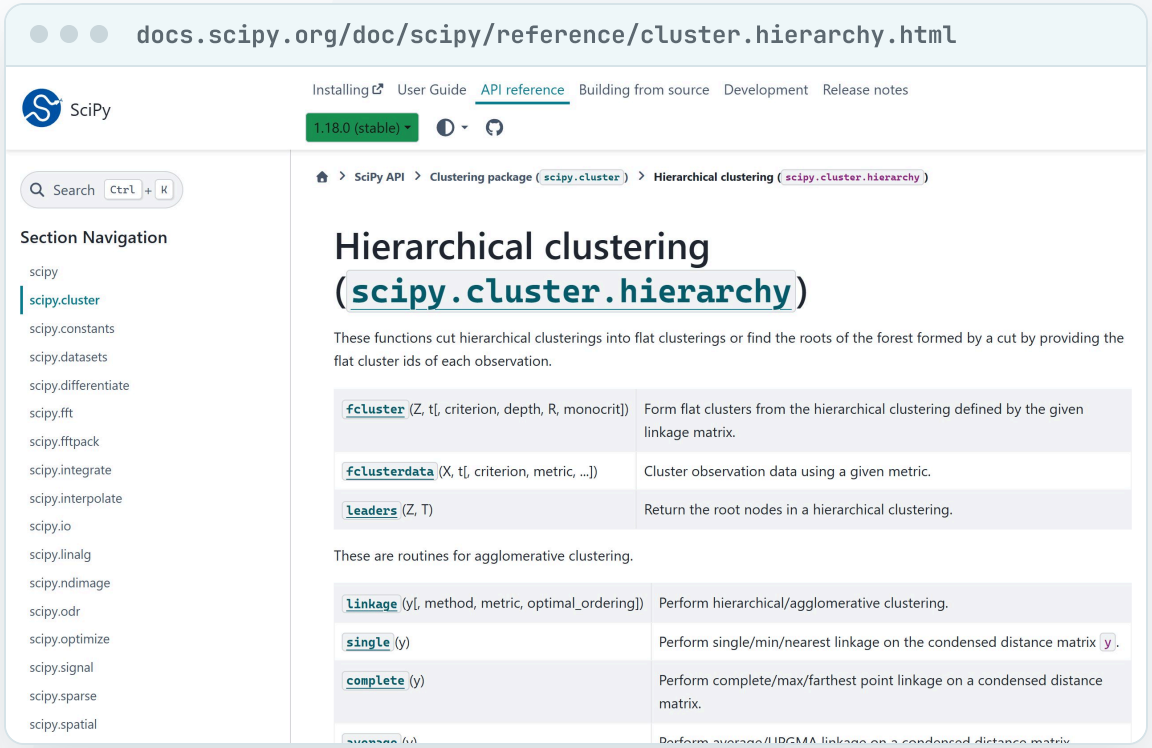
Medians, not means
one huge customer can drag a mean
far from the typical member

Check against what you know
the held-back channel shows whether a
segment matches a real kind of business

scikit-learn and SciPy Docs



scikit-learn: every clustering method side by side



SciPy: linkage and dendrogram

The scikit-learn guide opens with a grid of every method on the same toy datasets.

Choosing a Clustering Method

METHOD	YOU SET	CLUSTER SHAPE	NOISE	USE IT FOR
k-means	k	round, similar size	none	a fast first answer, large data
Agglomerative (Ward)	k or a cut height	compact	none	a tree to read; small data
DBSCAN	eps, min_samples	any	yes	even density, few dimensions
HDBSCAN	min_cluster_size	any	yes	uneven density, after UMAP
Gaussian mixture	k	ellipses	none	soft memberships, uncertainty

SECTION 04 · COMPLETE

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